Sri Lanka Institute of Information Technology

IE2100 – Data Communication and Wireless Network

Year 2, Semester 2 - 2026

Campus/ Center name- SLIIT/Malabe

Specialization - CSNE

Assignment 01

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Executive Summary

The purpose of this proposal is to provide a design for a scalable and secure wired/wireless network for Behpaya Information Technology that consists of a two-floored building located in Sri Lanka.

This solution has been designed to serve the approximate number of 50 staff members, 30 simultaneous guest devices, and overall 120 active endpoints in the network.

In order to achieve the goal, hierarchical enterprise network architecture based on the use of next generation firewall, multilayer core switch, floor-based managed access switches, and properly arranged wireless access points will be used in order to ensure wired and wireless coverage for Behpaya Information Technology.

Logical segmentation into VLANs in order to isolate administrative networks, staff, guests, management networks, and server resources is one of the key aspects of network design. Additionally, security measures such as VLANs segmentation, perimeter protection provided by firewall, support of virtual private network, and access control list enforcement will be used in order to enhance overall network security.

Network monitoring, configuration backups, and disaster recovery plans are covered in the proposed network management system architecture, while an estimated financial proposal is also presented in this paper. Finally, Cisco Packet Tracer simulation was used in order to test

1. Project Management and Proposal Approach

1.1 Project Overview

This project proposes a secure, reliable, and scalable wired and wireless network infrastructure for Behpaya Information Technology's two-floor office in Sri Lanka. The proposed solution is designed to support around 50 staff members, guest users, and up to 120 active endpoints while ensuring strong security, centralized management, wireless coverage, and future scalability for the next 3 to 5 years. The proposal includes wired LAN design, wireless LAN design, security architecture, IP and VLAN planning, monitoring, testing, and financial estimation. This project will be supported with technical documentation and simulation

1.2 Project Objectives

- Design a modern wired LAN for a two-floor office.
- Design a wireless LAN with proper AP placement and SSID planning.
- Provide VLAN-based segmentation for staff, admin, guest, and network devices.
- Improve network security using firewall, VPN, and access control.
- Prepare IP addressing, DHCP, DNS, and management plans.
- Create a financial proposal including hardware and implementation cost.
- Provide supporting diagrams and documentation

1.3 Project Methodology

Phase 1 – Requirement Analysis
The assignment requirements were studied carefully to identify the office layout, number of users, device capacity, network services, wireless needs, security expectations, and deliverables.

Phase 2 – Conceptual Network Planning
A high-level architecture was prepared to decide the core network components such as firewall, multilayer switch, access switches, access points, VLAN structure, and service layout.

Phase 3 – Detailed Design
The project team prepared the detailed wired LAN design, wireless LAN design, IP addressing scheme, VLAN plan, SSID mapping, and security policies.

Phase 4 – Simulation and Validation
The network topology was created in Cisco Packet Tracer to simulate VLAN segmentation, routing, DHCP, wireless connectivity, and testing.

Phase 5 – Documentation and Financial Proposal
Technical design details, diagrams, testing results, and cost estimations were compiled into the final technical and financial proposal.

1.4 Resource Allocation and Team Responsibilities

Member	Responsibility Area	Main Tasks
Member 1	Project Management and Documentation	Requirement analysis, methodology, timeline, risk management, report formatting, final proposal coordination
Member 2	Wired LAN and IP Design	Switch topology, VLAN design, subnetting, trunking, inter-VLAN routing, wired simulation
Member 3	Wireless Network Design	AP placement, SSID planning, wireless security, roaming concept, coverage planning
Member 4	Security, Monitoring, and Financial Proposal	Firewall design, VPN, threat analysis, NMS planning, backup policy, cost estimation

Table 1.1

1.5 Project Timeline

Phase	Activity	Duration
1	Requirement analysis and project planning	Day 1
2	High-level architecture design	Day 2
3	Wired LAN and VLAN design	Day 3
4	Wireless LAN and AP placement design	Day 4
5	Security and management design	Day 5
6	Packet Tracer simulation and testing	Day 6
7	Cost estimation and financial proposal	Day 7
8	Final report writing	Day 8
9	Slide preparation and viva practice	Day 9

Table 1.2

The project timeline was organized to ensure that planning, design, simulation, documentation, and presentation preparation were completed in a structured sequence. This helped reduce errors and made it easier to manage each technical component systematically.

1.6 Risk Management

Risk	Possible Impact	Mitigation
Incorrect requirement interpretation	Design may not match client needs	Recheck assignment brief before finalizing design
IP/VLAN planning errors	Connectivity problems in simulation	Review addressing plan before configuration
Too few access points	Weak wireless coverage	Plan AP placement based on office areas and user density
Guest access not isolated	Security issue	Use separate guest VLAN and internet-only access policy
Budget exceeds realistic level	Proposal may appear impractical	Choose cost-effective enterprise devices and state assumptions
Packet Tracer limitations	Some enterprise features cannot be fully simulated	Explain advanced features conceptually in report
Time management issues	Delayed completion	Follow a clear phased schedule

Table 1.3

1.7 Communication and Reporting Plan

The project team communicates through regular discussion during the planning, design, and documentation stages. Progress is reviewed phase by phase to ensure that each design area aligns with the client requirements. Draft outputs such as diagrams, simulation files, cost tables, and report sections are checked before final integration.

1.8 Project Deliverables

- Technical proposal for the office network
- Financial proposal with estimated cost
- High-level and low-level network diagrams
- VLAN and IP addressing plan
- Wireless AP placement and SSID design
- Security architecture and management plan
- Cisco Packet Tracer simulation
- Testing and validation screenshots

2. Network Architecture and Wired LAN Design

2.1 High-Level Network Architecture

The proposed network for the **Behpaya Information Technology office** is based on hierarchical **network architecture**, designed to ensure scalability, security, efficient traffic management, and ease of administration for a two-floor office environment.

The network connects to the internet through an **edge firewall/router**, which serves as the primary security gateway. This device provides internet access, enforces perimeter security policies, supports VPN connectivity, and filters inbound and outbound traffic.

At the core of the network, a **multilayer core switch** located in the Main Distribution Frame handles inter-VLAN routing, acts as the default gateway for all VLANs, and aggregates traffic from both floors. This ensures high-speed, reliable communication across the organization.

Each floor is equipped with a **managed access switch** that provides wired connectivity for end-user devices, printers, and wireless access points. These access switches connect to the core switch using trunk links, allowing secure and efficient VLAN traffic flow between floors.

Wireless access points are deployed on both floors to deliver full Wi-Fi coverage. Multiple SSIDs are mapped to separate VLANs to securely support corporate and guest wireless users. Overall, this hierarchical design enhances performance, improves fault isolation, simplifies troubleshooting, and supports future growth, aligning with enterprise best practices for small-to-medium business networks.

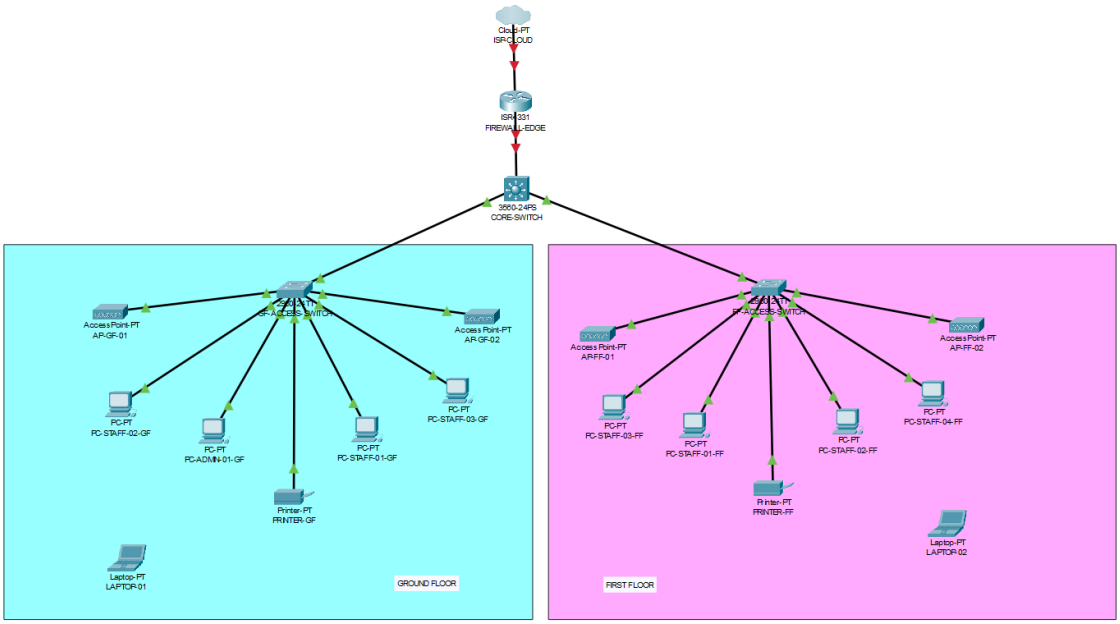


Figure 2.1: Topology

2.2 Proposed Device Inventory and Justification

Enterprise level hardware is going to be used to support the security requirements, wireless connection, and scalability of the network infrastructure being proposed. The edge firewall/router serves as a gateway to the internet, offers security services, and acts as a virtual private network (VPN) and filter. The core switches are multilayer switches that provide VLAN routing capability. Access switches support wired devices and wireless access points per floor, while also strategically placing wireless access points to ensure Wi-Fi roaming capability on both floors.

2.2.1 Infrastructure Devices

Device	Quantity	Purpose	Justification
Router / Firewall	1	WAN Connectivity & Security	Provides internet access, firewall filtering, VPN, perimeter security
Layer 3 Core Switch	1	Core Distribution	Performs inter-VLAN routing and central switching
Access Switch	2	Floor Access Layer	Connects wired devices on each floor
Wireless Access Point	4	Wireless Coverage	Provides office-wide Wi-Fi coverage
Network Printer	2	Shared Printing	One printer per floor for convenience

Table 2.1: Infrastructure Device Justification Table

2.2.2 End-User Device

Device Type	Estimated Quantity	Purpose
Desktop PCs	50 staff users (represented partially in simulation)	Staff and administrative workstations
Laptops	Included within staff and wireless user environment	Mobile office access
Printers	2	Shared departmental printing
Guest Devices	Up to 30 concurrent devices	Visitor internet-only access
Total Active Endpoints	Up to 120	Full office network capacity

Table 2.2: End-User Device Justification Table

Note:
The Packet Tracer simulation includes a reduced number of end devices for demonstration and validation purposes. However, the logical design, VLAN structure, addressing plan, and wireless architecture were developed to support the full client requirement of approximately 50 staff, 30 guest devices, and up to 120 active endpoints.

2.3 VLAN Design and Segmentation Plan

The proposed network uses VLAN-based segmentation to logically separate traffic between different departments, services, and user groups. VLAN segmentation improves security, reduces broadcast domains, enhances traffic management, and simplifies policy enforcement across the enterprise network.

Separate VLANs are assigned for administrative users, general staff, guest wireless users, network management devices, printers, and future IoT expansion. This design ensures that sensitive administrative traffic remains isolated from regular staff and guest traffic while allowing centralized inter-VLAN routing through the core multilayer switch.

The VLAN structure has been designed in alignment with the client requirement for segmentation between staff, administration, guests, and network devices while maintaining future scalability.

VLAN ID	VLAN Name	Purpose	Associated Devices
10	ADMIN	Administrative / IT Users	Admin PCs
20	STAFF	General Employee Devices	Staff PCs, Corporate Wireless
30	GUEST	Guest Wireless Network	Guest Wireless Devices
40	MGMT	Network Management	Switches, AP Management
50	SERVER	Shared Services / Printers	Printers, Internal Server
60	IOT	Future Expansion / IoT	Reserved for Future Use

Table 2.3: VLAN Design Table

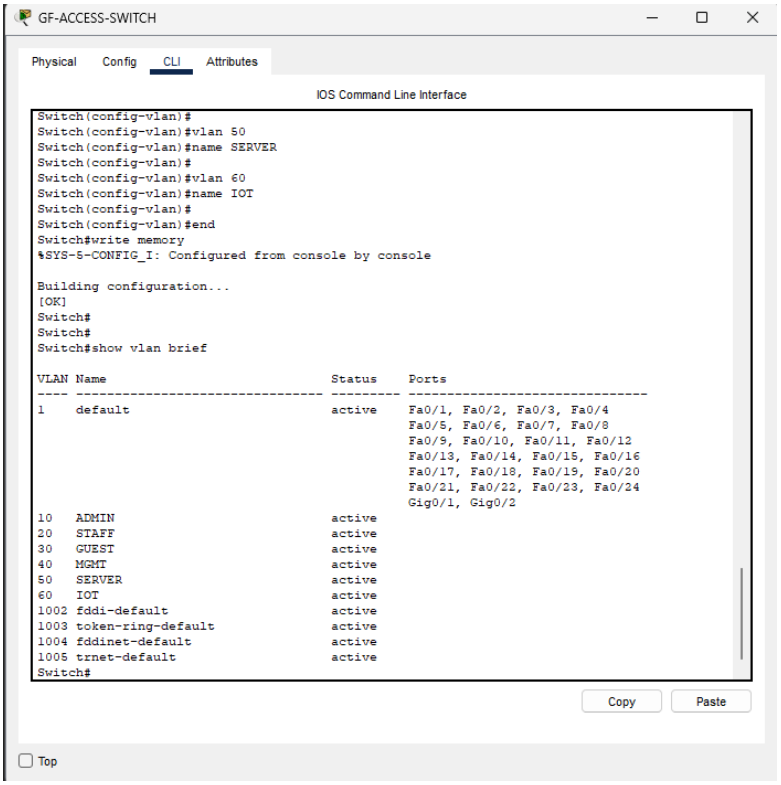


Figure 2.2: VLAN Creation Verification

2.4 IP Addressing Scheme

A structured private IP addressing scheme was developed for the proposed network using Class C private addressing ranges. Each VLAN is assigned a dedicated /24 subnet to simplify management, subnetting, troubleshooting, and future scalability.

The multilayer core switch acts as the default gateway for each VLAN through Switched Virtual Interfaces (SVIs). This allows centralized inter-VLAN routing while maintaining subnet separation.

VLAN ID	Subnet	Default Gateway	DHCP Scope Range
10	192.168.10.0/24	192.168.10.1	192.168.10.50 – 192.168.10.200
20	192.168.20.0/24	192.168.20.1	192.168.20.50 – 192.168.20.200
30	192.168.30.0/24	192.168.30.1	192.168.30.50 – 192.168.30.200
40	192.168.40.0/24	192.168.40.1	192.168.40.50 – 192.168.40.200
50	192.168.50.0/24	192.168.50.1	192.168.50.50 – 192.168.50.200
60	192.168.60.0/24	192.168.60.1	192.168.60.50 – 192.168.60.200

Table 2.4: IP Addressing Table

Note:
The DHCP scope ranges reserve lower IP addresses for static assignment of gateways, infrastructure devices, servers, printers, and network management interfaces.

2.5 Inter-VLAN Routing Explanation

Inter-VLAN routing was implemented on the multilayer core switch using Switched Virtual Interfaces (SVIs). Each VLAN was assigned a dedicated gateway IP address, allowing centralized routing between VLANs while maintaining logical segmentation.

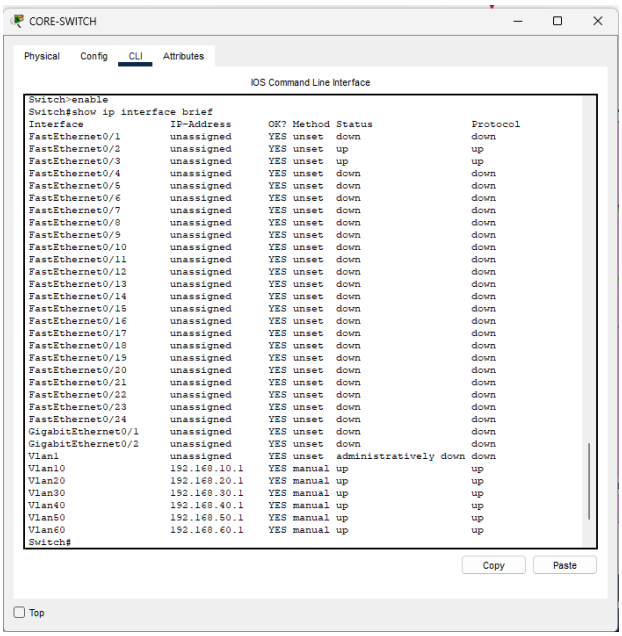
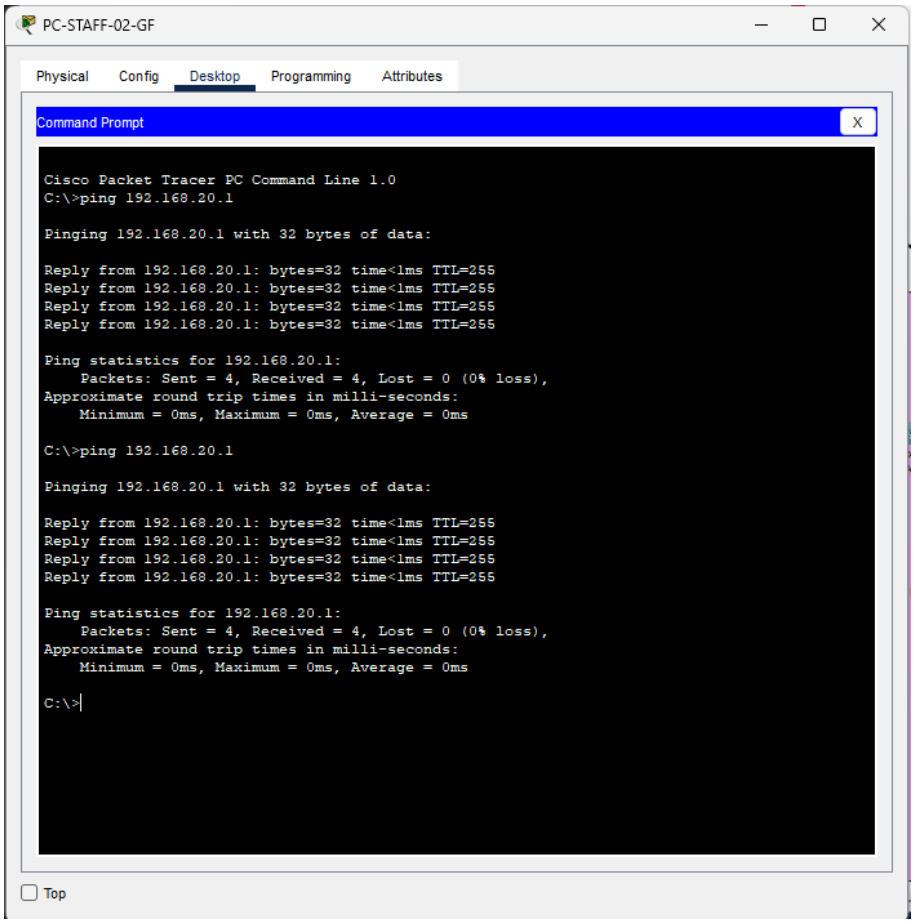


Figure 2.3: SVI Configuration

2.6 Packet Tracer Simulation and Validation

The implementation and testing of the wired network were done through the use of Cisco Packet Tracer. VLAN segregation, trunking, inter-VLAN routing, IP addressing for end devices and printers, and DHCP were all part of the simulation process. The core multi-layer switch managed the SVIs and the inter-VLAN routing, whereas the access layer switches offered VLAN connectivity spanning across two floors. End devices utilized DHCP, and the printers had static IP addresses.

Figure 2.4: Gateway Ping



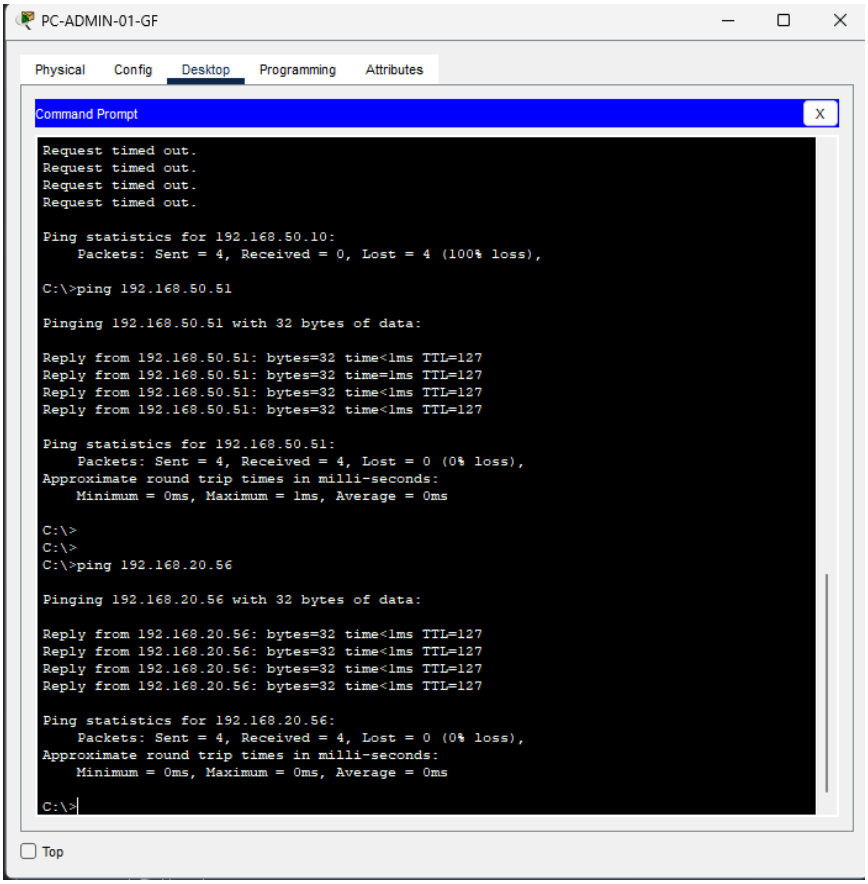


Figure 2.5: Inter VLAN Ping

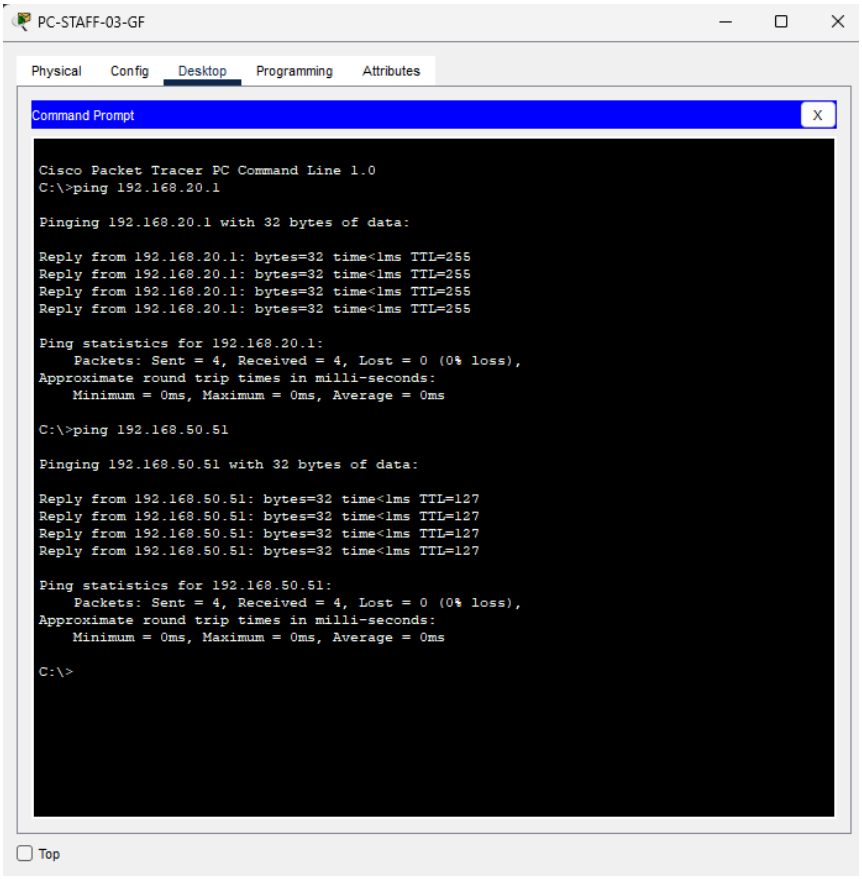


Figure 2.6: Printer Ping

Test Item	Expected Result	Outcome
VLAN creation	VLANs available on all switches	Successful
Trunk links	VLAN traffic passes between switches	Successful
Access port assignment	Devices join correct VLANs	Successful
SVI gateway configuration	VLAN gateways reachable	Successful
DHCP allocation	Clients receive correct IP settings	Successful
Printer static IP configuration	Printers reachable with fixed IPs	Successful
Inter-VLAN routing	Devices in different VLANs can communicate as permitted	Successful

Table 2.5: Testing

3. Wireless Network Design

3.1 WLAN Coverage and AP Placement

The wireless local area network has been set up to achieve consistent Wi-Fi connectivity throughout the entire two-story office building. There are four wireless access points, with two per floor, strategically installed to guarantee consistent connectivity, easy roaming capability, and zero dead spots in the entire office building. The objective of the setup is to attain a minimum signal strength of −65 dBm.

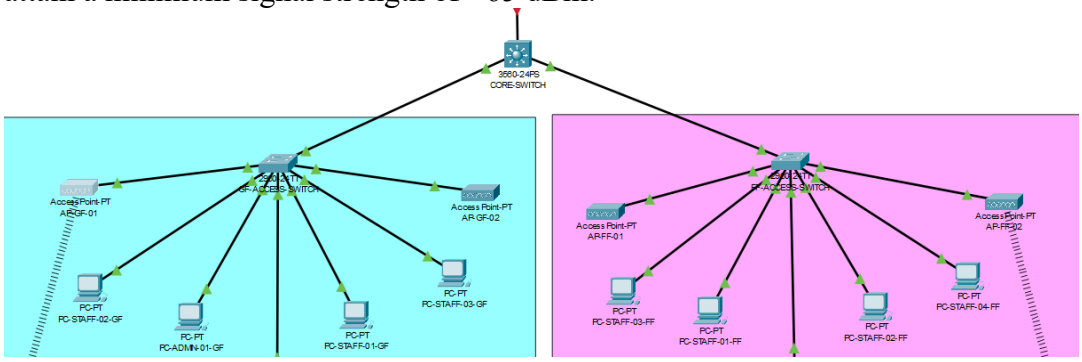


Figure 3.1: AP Placement

3.2 SSID Design and VLAN Mapping

Separate wireless SSIDs are used to logically segment employee and guest wireless traffic. Corporate wireless users are mapped to the staff VLAN, while guest users are placed in an isolated guest VLAN to ensure internet-only access and prevent lateral movement into internal corporate resources.

SSID	VLAN	Purpose
Behpaya-Corp	20	Corporate Staff Wireless Access
Behpaya-Guest	30	Guest Internet-Only Access

Table 3.1: SSID Mapping Table

3.3 Wireless Security and Authentication

Corporate wireless access is secured using WPA2/WPA3-Enterprise authentication principles to ensure secure user authentication and encrypted wireless communication. Guest wireless access is logically separated through VLAN isolation and is intended to support captive portal/rate-limited internet-only access in a production deployment.

Due to Packet Tracer simulation limitations, advanced enterprise wireless authentication and captive portal features are represented conceptually in the design documentation while simplified wireless security settings are used in the simulation environment.

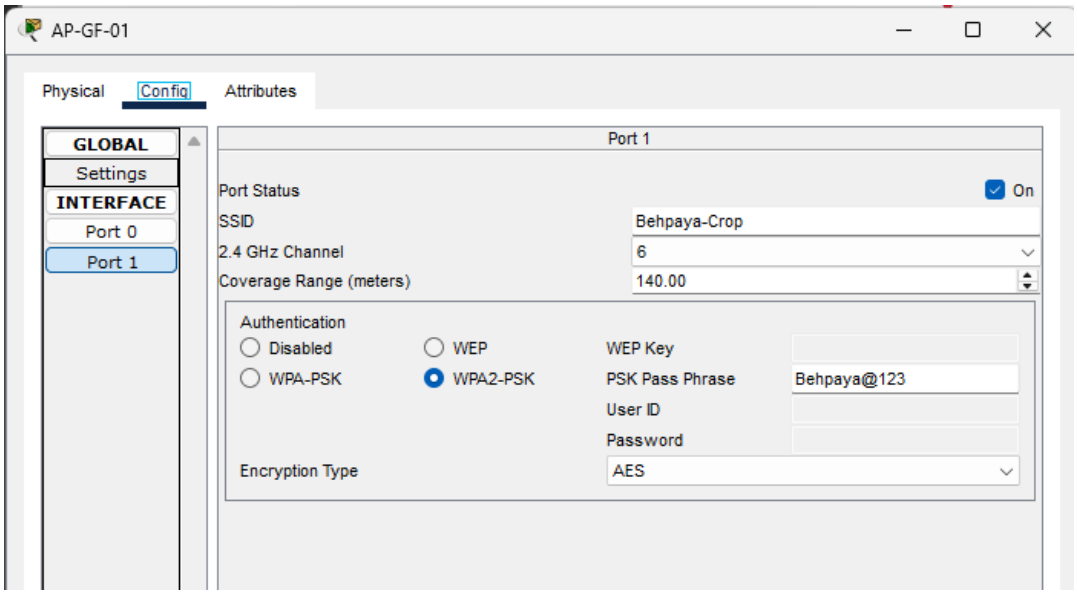


Figure 3.1: AP Configuration

3.4 Wireless Simulation and Validation

Wireless connectivity was simulated in Cisco Packet Tracer by configuring all access points with a common corporate SSID and WPA2-PSK security settings. Wireless client devices successfully associated with the configured SSID and authenticated using the configured passphrase.

Connected wireless clients were able to obtain IP configuration through DHCP and successfully communicate with internal network resources, validating the wireless design and confirming operational WLAN functionality within the simulated environment.

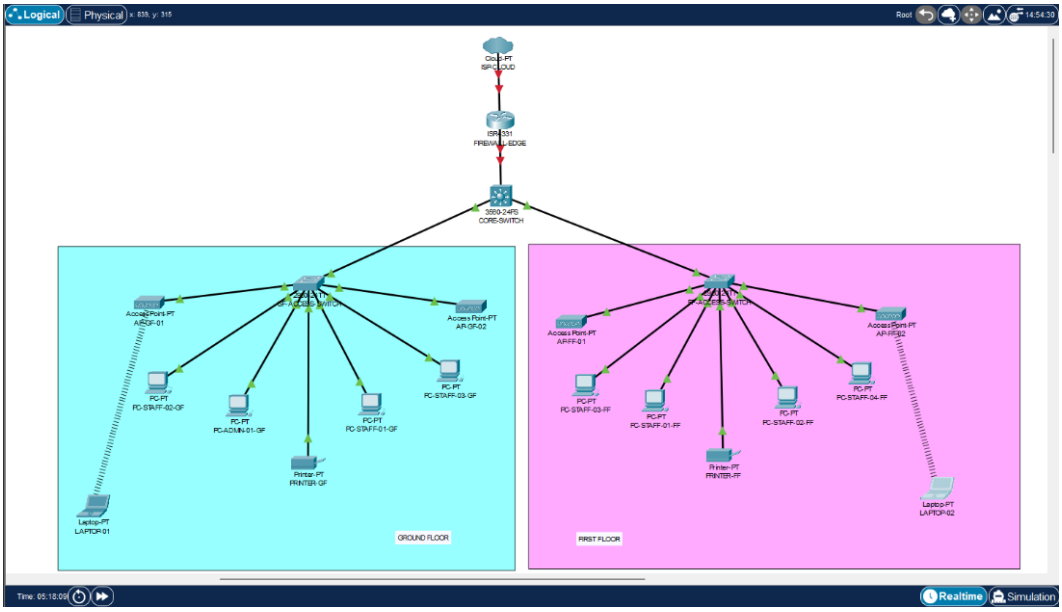


Figure 3.2: Wireless client connection

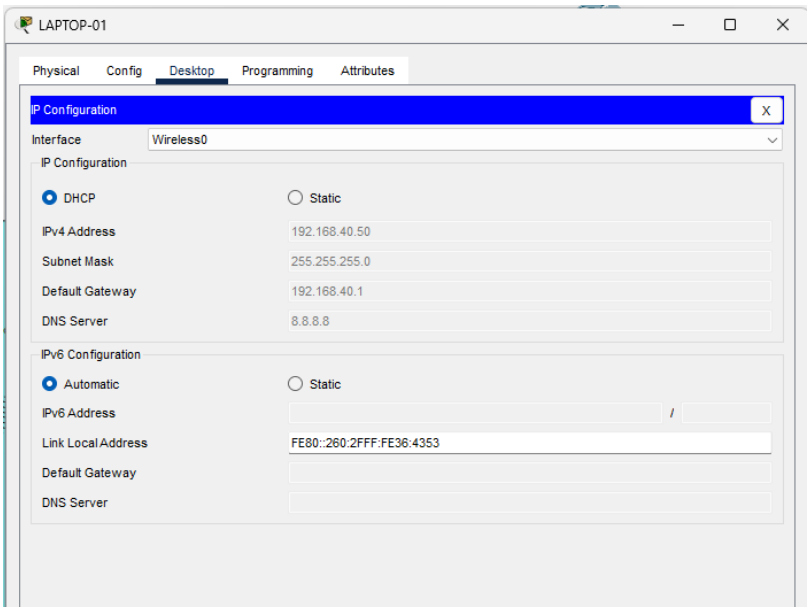


Figure 3.3: Wireless DHCP/IP

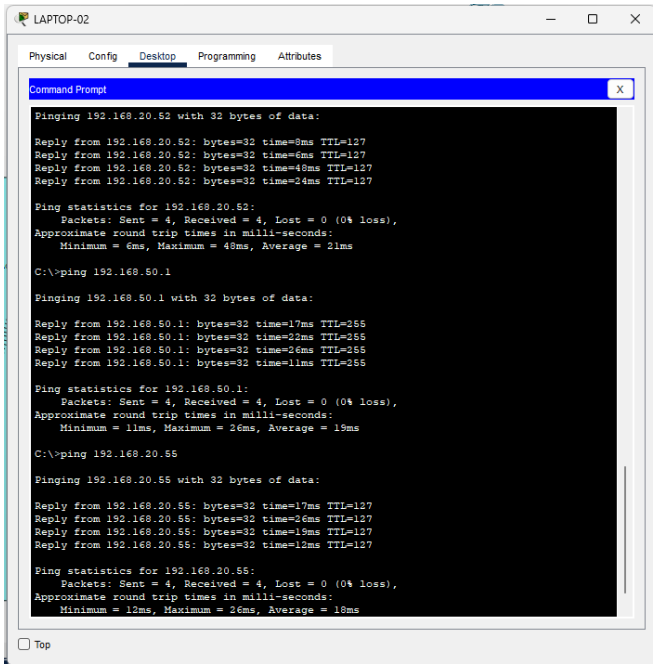


Figure 3.4: Successful Wireless Ping Test

4. Security Design

4.1 Threat Identification

The proposed network design considers several common enterprise security threats that could impact the confidentiality, integrity, and availability of the organization's network infrastructure and business operations.

Potential threats include unauthorized internal access, malware propagation, guest-to-corporate lateral movement, internet-based attacks, credential theft, denial-of-service attacks, and unauthorized device connections.

A layered security approach is therefore incorporated into the proposed design to reduce risk and enforce secure access policies across the wired and wireless infrastructure.

Threat	Risk to Organization	Mitigation
Unauthorized internal access	Exposure of sensitive data	VLAN segmentation, RBAC
Guest network abuse	Access to internal network	Guest VLAN isolation
Malware / malicious websites	Endpoint compromise	Firewall AV/URL filtering
External internet attacks	Network breach	NGFW / IPS / Geo-IP filtering
Credential compromise	Unauthorized login	MFA / WPA2/WPA3 / VPN
Rogue devices	Unauthorized network access	NAC / Port security / 802.1X
Device failure / misconfiguration	Downtime	Backup configs / DR plan

Table 4.1

4.2 Security Architecture and Implementation

A defense-in-depth security model is proposed for the Behpaya IT office network. A next-generation firewall is positioned at the network edge to provide perimeter protection, traffic inspection, VPN termination, and policy enforcement between internal VLANs and external networks.

Internal security is enhanced through VLAN-based segmentation, isolating administrative, staff, guest, management, and server resources into dedicated logical networks. This segmentation reduces attack surface and limits lateral movement between departments and device classes.

Guest wireless traffic is isolated in a dedicated guest VLAN and intended to be restricted to internet-only access through firewall policy enforcement. Administrative access to network devices is restricted to authorized users within the administration VLAN.

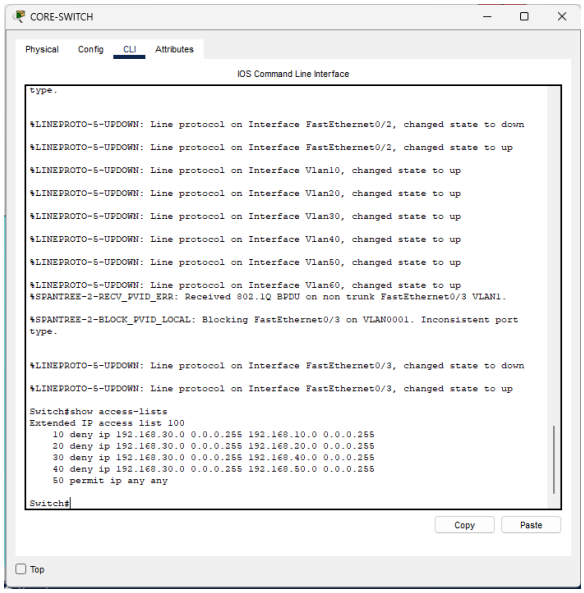


Figure 4.1: ACL Configuration

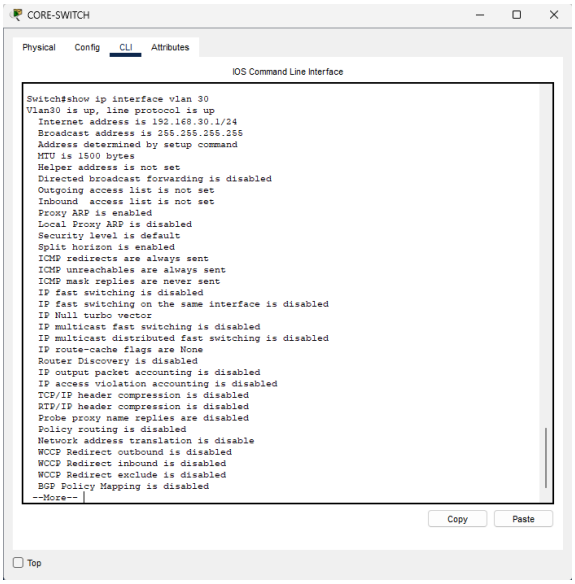


Figure 4.2: ACL Applied to VLAN 30 Interface

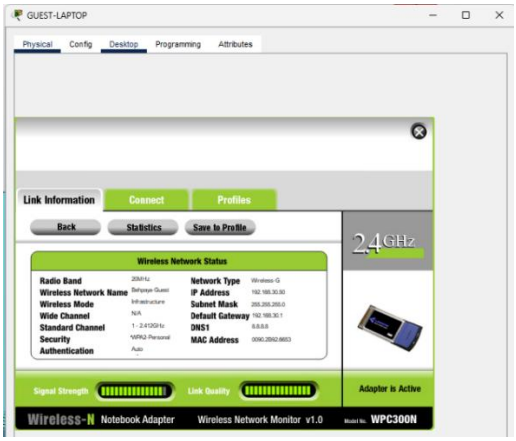


Figure 4.3: Guest Laptop Connected to Guest Network

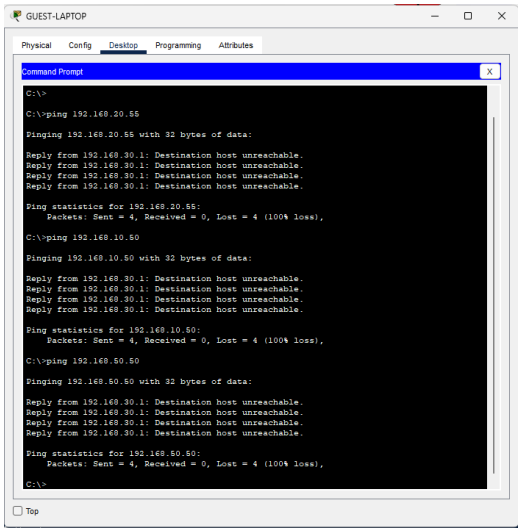


Figure 4.4: Failed Ping

Security Validation

Validation testing confirmed that guest clients assigned to VLAN 30 were unable to access internal administrative, staff, management, and printer VLANs while remaining logically isolated from corporate resources. This confirms successful implementation of guest network segmentation and ACL-based access control.

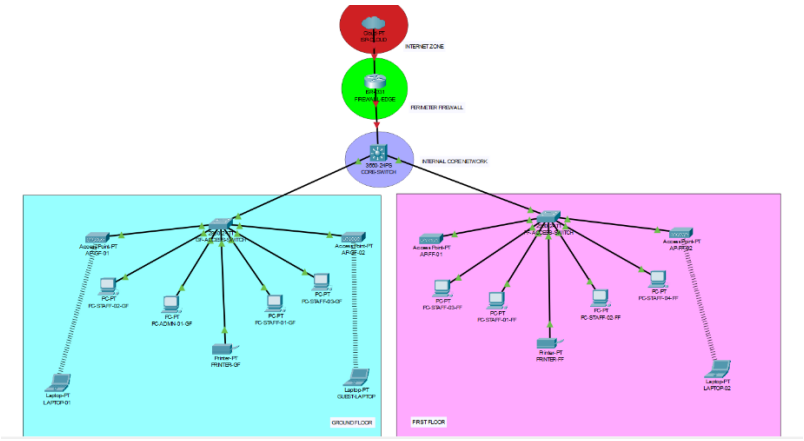


Figure 4.5: Security Architecture

4.3 VPN and Remote Access Design

Secure remote connectivity is supported through VPN services terminated at the edge firewall. Remote employees and administrators can connect securely using SSL/IPsec VPN technologies with multi-factor authentication in a production deployment.

VPN access is intended to be role-based and limited according to user privileges to ensure secure remote administration and remote workforce support.

4.4 Access Control and Segmentation Policy

Network segmentation policies are implemented to control communication between VLANs and enforce least-privilege access principles.

Administrative VLAN users are granted privileged access to management and infrastructure resources. Staff VLAN users may access internal business resources as permitted by policy. Guest VLAN users are restricted from accessing internal corporate resources and are limited to internet-only connectivity.

These controls improve internal network security and align with enterprise segmentation best practices.

5. Network Management System Design

5.1 Proposed NMS Solution

To support centralized monitoring and operational visibility, a Network Management System (NMS) is proposed for the Behpaya IT office network. The recommended solution for this deployment is a centralized monitoring platform such as PRTG Network Monitor, Zabbix, or SolarWinds, depending on organizational budget and operational preferences.

The NMS platform will provide centralized monitoring of switches, access points, firewall status, WAN connectivity, bandwidth utilization, and device health through SNMP and telemetry-based monitoring mechanisms.

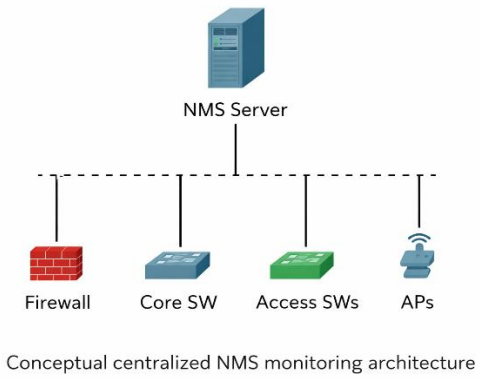


Figure 5.1: NMS Diagram

5.2 Monitoring and Alerting Strategy

Threshold-based monitoring and alerting are proposed to proactively detect infrastructure issues and reduce operational downtime. Alerts can be generated for WAN outages, access point failures, switch uplink failures, excessive interface errors, CPU/memory utilization thresholds, and abnormal traffic conditions.

This proactive monitoring approach improves incident response time and operational reliability.

Monitoring Item	Alert Condition	Purpose
WAN Link	Link Down	Detect ISP/Internet outage
Firewall	Offline / High CPU	Detect perimeter issues
Core Switch	Port/Uplink Failure	Detect switching faults
Access Points	Offline	Detect WLAN outage
Interface Errors	Threshold Exceeded	Detect cable/device issues
Bandwidth Usage	High Utilization	Capacity monitoring

Table 5.1: Monitoring Table

5.3 Configuration Backup and Disaster Recovery

Regular automated backups of network device configurations are proposed to ensure rapid restoration in the event of hardware failure, misconfiguration, or disaster scenarios. Configuration backups should be securely stored on internal backup servers or centralized management platforms.

A disaster recovery procedure should include restoration of device configurations, replacement of failed hardware, verification of connectivity, and staged recovery of network services.

5.4 Administrative Access Control

Administrative access to network infrastructure should be controlled using role-based access control (RBAC), restricting management privileges to authorized personnel only. Administrative authentication may be integrated with centralized identity platforms where available, while local fallback credentials should be securely maintained for emergency access.

Management access should be restricted to trusted administrative VLANs and encrypted protocols such as SSH/HTTPS.

6. Financial Proposal

A financial estimate was prepared for the proposed network solution based on the required infrastructure components, wireless equipment, security appliances, structured cabling, and implementation services. Pricing values represent estimated market-based academic approximations and are intended for proposal evaluation purposes only.

The financial proposal includes hardware procurement, software/licensing assumptions, structured cabling materials, and deployment/implementation labor.

6.1 Bill of Materials (BOM)

Item	Qty	Unit Cost (LKR)	Total Cost (USD)	Justification
Next-Generation Firewall	1	354,300	354,300	Perimeter security / VPN / Internet edge
Layer 3 Core Switch (48 Port)	1	220,000	220,000	Core routing and aggregation
Access Switch (48 Port Managed)	3	110,650	331,950	Supports ~120 wired/AP uplink ports with growth
Enterprise Wireless Access Point	4	42,350	169,400	Office-wide wireless coverage TP-LINK CPE510 5G 300M
Internal Server / NMS Platform	1	500,00	500,000	Monitoring / Backup / Services
Rack Cabinet 12U	1	25,450	25,450	Equipment housing
Patch Panels	3	2,400	7,200	Structured cabling termination
Cat6 Structured Cabling	1 box	69,000	69,000	Full office cabling deployment
DIP 650VA UPS	1	9,250	9,250	Power protection
Sonic wall Network Security Manager	1 Lot	187,990	187,990	Firewall/NMS licensing
Installation / Configuration Labor	12	5,000	60,000	Professional deployment services

Table 6.1: Network Infrastructure Cost Assumptions

Total Amount: LKR. 1,934,540

6.2 Cost Estimation

Item	Qty	Unit Cost (LKR)	Total Cost (LKR)	Purpose
Staff Desktop PCs DELL I3 14th Gen	50	250,500	12,525,000	Employee Workstations
Staff Desktop PC monitors MSI PRO MP251L	50	37,560	1,878,000	Employee Workstations
Staff Laptops DELL VOSTRO 3520	10	146,000	1,460,000	Mobile/Executive Staff
Guest/User BYOD Devices	Client-Owned	N/A	N/A	Not Included in Budget
Network Printers HP Smart Tank 210	2	51,000	102,000	Shared Printing

Table 6.2: Endpoint Device Cost Assumptions

Total Amount: LKR. 15,965,000

Final Total Amount = 1,934,540 + 15,965,000 = **LKR. 17,899,540 /=**

6.3 Budget Summary and Assumptions

The above pricing values are estimated market-based figures from **sense.lk** .
Budget Could be increased due to government taxes and Other extra Expenses.

Conclusion

The proposed network design provides Behpaya Information Technology with a secure, scalable, and future-ready wired and wireless networking solution aligned with the organization's operational and technical requirements.

The design successfully addresses the client's needs for reliable LAN and WLAN connectivity, logical traffic segmentation, enterprise-grade security controls, centralized monitoring, and support for future growth over the next three to five years.

Simulation and validation performed using Cisco Packet Tracer confirmed the functionality of the logical network design, including VLAN segmentation, inter-VLAN routing, DHCP services, wireless connectivity, and guest access control enforcement.

Overall, the proposed solution delivers a balanced combination of performance, security, manageability, and cost-effectiveness suitable for a modern small-to-medium enterprise office environment.